Symmetric Tucker Tensor Decomposition

1 Introduction

Let

$$\mathcal{X} \in S^d(\mathbb{R}^n)$$

be an order-d symmetric tensor, where $S^d(\mathbb{R}^n)$ denotes the space of d-way symmetric tensors over $\mathbb{R}^n$.

A standard Tucker decomposition of a tensor has the form

$$\mathcal{X} \approx \mathcal{G} \times_1 U^{(1)} \times_2 U^{(2)} \cdots \times_d U^{(d)},$$

where each mode can have a different factor matrix.

In a *symmetric Tucker decomposition*, we enforce the same factor matrix in every mode. Thus the approximation becomes

$$\mathcal{X} \approx \mathcal{C} \times_1 Q \times_2 Q \cdots \times_d Q,$$

where

$$Q \in \mathbb{R}^{n \times r}, \qquad Q^\top Q = I_r,$$

and the core tensor

$$\mathcal{C} \in S^d(\mathbb{R}^r)$$

is itself symmetric.

So the key idea is simple: instead of learning a different subspace for each mode, we learn one shared subspace Q that is used across all modes.

2 Notation

We will use the following notation throughout:

- $\mathcal{X}$: the original symmetric tensor.
- $\mathcal{C}$: the symmetric Tucker core.
- $Q \in \mathbb{R}^{n \times r}$: the shared factor matrix with orthonormal columns.
- $\times_k$: mode-k tensor-matrix product.
- $\|\cdot\|$: Frobenius norm.

The symmetric Tucker model is therefore

$$\boxed{\mathcal{X} \approx \mathcal{C} \times_1 Q \times_2 Q \cdots \times_d Q}$$

with the same matrix Q repeated d times.

3 Optimization Problem

The best rank-r symmetric Tucker approximation is obtained by solving

$$\min_{\mathcal{C}, Q} \|\mathcal{X} - \mathcal{C} \times_1 Q \times_2 Q \cdots \times_d Q\|^2 \qquad \text{subject to} \qquad Q^\top Q = I_r,$$

with $\mathcal{C} \in S^d(\mathbb{R}^r)$.

This means we want to find

1. a shared r-dimensional subspace Q, and
2. a smaller symmetric core tensor $\mathcal{C}$,

so that the reconstruction error is as small as possible.

4 Optimal Core for Fixed Q

If Q is fixed, then the optimal core is obtained by projecting the tensor onto the shared subspace in every mode:

$$\boxed{\mathcal{C} = \mathcal{X} \times_1 Q^\top \times_2 Q^\top \cdots \times_d Q^\top}$$

So once Q is known, the core is determined immediately.
Substituting this back into the objective gives

$$\mathcal{X}_{\text{approx}} = \mathcal{C} \times_1 Q \times_2 Q \cdots \times_d Q.$$

Hence the whole problem reduces to choosing the best shared factor matrix Q.

5 Equivalent Maximization Form

After eliminating the core, the problem becomes equivalent to maximizing the norm of the projected tensor:

$$\max_{Q^\top Q = I_r} \left\| \mathcal{X} \times_1 Q^\top \times_2 Q^\top \cdots \times_d Q^\top \right\|^2.$$

Define

$$F(Q) := \left\| \mathcal{X} \times_1 Q^\top \times_2 Q^\top \cdots \times_d Q^\top \right\|^2.$$

Then the optimization problem is

$$\boxed{\max_{Q^\top Q = I_r} F(Q)}$$

This viewpoint is very important:

> We are looking for the orthonormal matrix Q that captures as much of the tensor's energy as possible in a shared low-dimensional subspace.

6 Why This Is Symmetric Tucker

In ordinary Tucker decomposition, one would write

$$\mathcal{X} \approx \mathcal{G} \times_1 U^{(1)} \times_2 U^{(2)} \cdots \times_d U^{(d)}.$$

In symmetric Tucker decomposition, we impose

$$U^{(1)} = U^{(2)} = \cdots = U^{(d)} = Q.$$

Therefore,

$$\mathcal{X} \approx \mathcal{C} \times_1 Q \times_2 Q \cdots \times_d Q.$$

Because the same basis is used in every mode, the core tensor $\mathcal{C}$ remains symmetric.

So the symmetry enters in two places:

1. the original tensor $\mathcal{X}$ is symmetric,
2. the factor matrices are forced to be identical.

7 HOEVD Initialization

A natural initialization is the higher-order eigenvalue decomposition (HOEVD), which is the symmetric analogue of HOSVD.

First, matricize the tensor $\mathcal{X}$ along one mode. Since $\mathcal{X}$ is symmetric, all mode unfoldings carry equivalent information. Let $\mathrm{mat}(\mathcal{X})$ denote such a matricization.

Then form the Gram matrix

$$M = \mathrm{mat}(\mathcal{X})\,\mathrm{mat}(\mathcal{X})^\top.$$

Compute the top r eigenvectors of M, and place them as columns of Q:

$$Q_{\mathrm{HOEVD}} = \begin{bmatrix} q_1 & q_2 & \cdots & q_r \end{bmatrix}.$$

This gives an initial shared factor matrix.

After that, compute the corresponding core:

$$\mathcal{C}_{\mathrm{HOEVD}} = \mathcal{X} \times_1 Q_{\mathrm{HOEVD}}^\top \times_2 Q_{\mathrm{HOEVD}}^\top \cdots \times_d Q_{\mathrm{HOEVD}}^\top.$$

So HOEVD gives a direct approximation:

$$\mathcal{X} \approx \mathcal{C}_{\mathrm{HOEVD}} \times_1 Q_{\mathrm{HOEVD}} \times_2 Q_{\mathrm{HOEVD}} \cdots \times_d Q_{\mathrm{HOEVD}}.$$

This is often used as an initialization for a later iterative refinement step.

8 Projected Gradient Method

To improve over the HOEVD initialization, we optimize

$$F(Q) = \left\| \mathcal{X} \times_1 Q^\top \times_2 Q^\top \cdots \times_d Q^\top \right\|^2$$

subject to

$$Q^\top Q = I_r.$$

This is an optimization problem:

$$\mathrm{St}(n, r) = \{Q \in \mathbb{R}^{n \times r} : Q^\top Q = I_r\}.$$

8.1 Gradient Step

Let Q_t be the current iterate. We compute the gradient of $F(Q)$, denoted $\nabla F(Q_t)$, and take a step:

$$\widetilde{Q}_{t+1} = Q_t + \gamma_t \nabla F(Q_t),$$

where $\gamma_t > 0$ is the step size.

8.2 Re-orthonormalization Step

The matrix $\widetilde{Q}_{t+1}$ will generally not satisfy

$$\widetilde{Q}_{t+1}^\top \widetilde{Q}_{t+1} = I_r.$$

So we re-orthonormalize it using QR factorization:

$$\widetilde{Q}_{t+1} = \widehat{Q}_{t+1} R_{t+1},$$

and then set

$$Q_{t+1} = \widehat{Q}_{t+1}.$$

This ensures that every iterate stays on the constraint set

$$Q_{t+1}^\top Q_{t+1} = I_r.$$

8.3 Core Update

Once the iteration converges to a final factor matrix Q, we compute the symmetric core by projection:

$$\boxed{\mathcal{C} = \mathcal{X} \times_1 Q^\top \times_2 Q^\top \cdots \times_d Q^\top}$$

Then the final approximation is

$$\boxed{\mathcal{X} \approx \mathcal{C} \times_1 Q \times_2 Q \cdots \times_d Q}$$

9 Algorithm

The full symmetric Tucker procedure can now be written clearly.

Input

A symmetric tensor

$$\mathcal{X} \in S^d(\mathbb{R}^n)$$

and a target rank r.

Output

A shared factor matrix

$$Q \in \mathbb{R}^{n \times r}, \qquad Q^\top Q = I_r,$$

and a symmetric core

$$\mathcal{C} \in S^d(\mathbb{R}^r)$$

such that

$$\mathcal{X} \approx \mathcal{C} \times_1 Q \times_2 Q \cdots \times_d Q.$$

Steps

1. **HOEVD initialization:** Matricize $\mathcal{X}$, form

$$M = \mathrm{mat}(\mathcal{X})\,\mathrm{mat}(\mathcal{X})^\top,$$

and take the top r eigenvectors to obtain an initial Q_0.

2. **Projected optimization:** For $t = 0, 1, 2, \ldots$, repeat:

$$\widetilde{Q}_{t+1} = Q_t + \gamma_t \nabla F(Q_t),$$

then compute a QR factorization

$$\widetilde{Q}_{t+1} = Q_{t+1} R_{t+1},$$

and keep the orthonormal part Q_{t+1}.

3. **Stopping criterion:** Stop when the change in Q_t or the objective $F(Q_t)$ becomes sufficiently small.

4. **Compute the core:** After convergence, compute

$$\mathcal{C} = \mathcal{X} \times_1 Q^\top \times_2 Q^\top \cdots \times_d Q^\top.$$

5. **Reconstruct the approximation:**

$$\mathcal{X}_{\mathrm{symTucker}} = \mathcal{C} \times_1 Q \times_2 Q \cdots \times_d Q.$$

10 Compact Mathematical Summary

The entire method can be summarized in the following compact form:

$$\boxed{\min_{\mathcal{C},\, Q^\top Q = I_r} \|\mathcal{X} - \mathcal{C} \times_1 Q \times_2 Q \cdots \times_d Q\|^2}$$

For fixed Q,

$$\boxed{\mathcal{C} = \mathcal{X} \times_1 Q^\top \times_2 Q^\top \cdots \times_d Q^\top}$$

Thus the reduced problem is

$$\boxed{\max_{Q^\top Q = I_r} \left\| \mathcal{X} \times_1 Q^\top \times_2 Q^\top \cdots \times_d Q^\top \right\|^2}$$

and the final approximation is

$$\boxed{\mathcal{X} \approx \mathcal{C} \times_1 Q \times_2 Q \cdots \times_d Q}$$

11 Interpretation

Symmetric Tucker decomposition can be interpreted as follows:

1. Find one shared low-dimensional basis Q.
2. Project the tensor onto that basis in every mode.
3. Store the compressed information in a smaller symmetric core $\mathcal{C}$.
4. Reconstruct the tensor using the same basis in every mode.

So this is the symmetric analogue of Tucker decomposition in exactly the same sense that a symmetric matrix eigendecomposition uses one common eigenbasis rather than different left and right singular spaces.

12 Conclusion

The symmetric Tucker decomposition of a symmetric tensor

$$\mathcal{X} \in S^d(\mathbb{R}^n)$$

takes the form

$$\mathcal{X} \approx \mathcal{C} \times_1 Q \times_2 Q \cdots \times_d Q,$$

where Q is shared across all modes and $\mathcal{C}$ is symmetric.

The procedure is typically carried out by:

1. obtaining an initial shared factor matrix using HOEVD,
2. refining Q using an optimization method under the orthonormality constraint,
3. computing the symmetric core by projection.

This gives a structured low-rank approximation that respects the symmetry of the tensor.